

# claude-windows / 9f6357d8-0e35-491a-a830-052e094c7f37

## Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_b6ef6275-de5\agent-afe16add3ef0f816.jsonl`
- SHA-256: `e2b0272c27a7d537fcacf26cec26180729ba3e65581910b10aeb7258933801e84`
- Source modified: `2026-06-21T14:00:47+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_b6ef6275-de5`
- session_id: `9f6357d8-0e35-491a-a830-052e094c7f37`
```

## Transcript

### 1. user (2026-06-21T13:58:45.952Z)

Synthesize a PRIORITIZED next-steps roadmap from these per-track doc scouts. The session has recently driven the APPLIANCE/CUJ + LOCALIZATION track (in-UI bucket?reel B-P2 just shipped + screencast; staleness corrected). The droplet e2e is PARKED by the owner (GPU needs a subscription). The Cloud-Run/serving track is being scanned separately ? reference it, don't duplicate.

PER-TRACK SCOUTS: [{"track": "appliance/CUJ", "docsRead": ["C:/Users/avidu/Projects/khelsutra/docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md", "C:/Users/avidu/Projects/khelsutra/docs/DRY\_RUN\_CUJ.md", "C:/Users/avidu/Projects/khelsutra/docs/PER\_RALLY\_SELECTION.md", "C:/Users/avidu/Projects/khelsutra/docs/REQUEST\_ACCESS\_FORM.md", "C:/Users/avidu/Projects/khelsutra/docs/PORTABILITY.md", "C:/Users/avidu/Projects/khelsutra/NEXT\_STEPS.md"], "statusSummary": "The appliance/CUJ track spans 5 docs plus NEXT\_STEPS. SHIPPED: (1) The dry-run CUJ goal (DRY\_RUN\_CUJ.md) is largely DONE ? \$6 DoD boxes 1-3 are ? (Path A local en+hi reel-download done 2026-06-21; Path B/B-P2 in-UI gs://bucket->process->12 rallies->reel-download live e2e done 2026-06-21 with screencast). Only box 4 (option 3 ? same CUJ on the CPU droplet) remains, and the OWNER JUST PARKED it (droplet un-provisioned, ~3.8GB RAM, no engine checkout; a real GPU machine needs a subscription). The two engine bugs the runs surfaced are both fixed (engine #281 queued-job hang; engine #289 compile-from-bucket 500). (2) PER\_RALLY\_SELECTION (the SPA back-half the CUJ rides on) is M0/MVP COMPLETE ? P0-P5 ? (pick->query->select->compile->download on existing endpoints, verified e2e + CUJ replay). (3) REQUEST\_ACCESS\_FORM: P0-P1 ?, P2 ? (Worker+D1 built+tested, owner-deploy pending ? but NEXT\_STEPS says the form is already LIVE on Cloudflare with D1+email, so P2 status in REQUEST\_ACCESS\_FORM.md is stale). (4) PORTABILITY: design + proven (Cloudflare<->Node/SQLite VM adapter swap, provision/teardown scripts tested live) ? no open tracker rows. IN PROGRESS / NOT STARTED: the broader LOCAL\_APPLIANCE\_ARCHITECTURE (KhelSutra Studio) is still DRAFT/owner-gated ? \$9 tracker has only T0 ? and T12 ? (engine half done); T1-T11 are ? (none started); T13 (GPU host) and T14 (bucket-worker bridge) are ? owner-gated. Note the broader appliance is NOT the current active focus ? NEXT\_STEPS names 'Friendly & Multilingual' (FRIENDLY\_AND\_MULTILINGUAL.md) as the active owner directive, a separate track.", "nextSteps": [{"step": "Confirm the bucket->reel CUJ on the CPU droplet (Linux deploy target) ? from-scratch engine install or a proper deploy of the fixed engine (#281/#289), then replay the in-UI gs://->process->rallies->compile->download flow through the gate.", "sourceDoc": "DRY\_RUN\_CUJ.md \$6 DoD box 4 (option 3) + \$10/\$4.3 droplet reality", "gated": "Blocked (PARKED by owner ? droplet un-provisioned, ~3.8GB RAM, no GPU; a real GPU machine needs a subscription)", "effort": "medium"}, {"step": "T1 ? Build the L1 mock engine + capability fixtures + browser-replay CUJ scripts (the hermetic <1s 4-CUJ harness).", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md \$9 tracker T1 (depends T0; not gated)", "gated": "No", "effort": "medium"}, {"step": "T2 ? Appliance shell: Caddyfile + systemd units + firewall + edge auth + rotate the chat-shared GEMINI\_API\_KEY (P1).", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md \$9 tracker T2", "gated": "Owner (edge-auth choice \$10 Q1: Caddy basicauth vs cloudflared+Cloudflare Access)", "effort": "medium"}, {"step": "T3 ? L3 live posture test: assert raw :8000 / :9000/:9001/:9023 are refused from outside.", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md \$9 tracker T3 (depends T2; not gated)", "gated": "Blocked (on T2 + a deployed box)", "effort": "small"}, {"step": "T4 ? S1 ingest screen: local-upload wired (existing /api/upload/init|part|complete) + segmenter picker from

/api/capabilities.", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 tracker T4 (depends T1; not gated)", "gated": "No", "effort": "medium"}, {"step": "T5 ? S2 progress screen: 3s poll of free-text stage + 'one job at a time' notice + restart-failure terminal state.", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 tracker T5 (depends T4)", "gated": "No", "effort": "small"}, {"step": "T6 ? S3: embed RallyPicker verbatim + compile + download (per-rally is one console screen).", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 tracker T6 (depends T4)", "gated": "No", "effort": "small"}, {"step": "T7 ? (engine) GET /api/capabilities + GET /api/health + UI capability banner. Note GET /api/health already EXISTS (engine M1/#268) per the 2026-06-21 correction, so the remaining net-new is /api/capabilities + the banner. (= PER\_RALLY\_SELECTION P6 family.)", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 T7 + PER\_RALLY\_SELECTION.md §7 P6", "gated": "Owner (coord with active engine sessions)", "effort": "medium"}, {"step": "T8 ? (engine) GET /api/videos + GET /api/reels/{video\_id} for S4 reload-survival (only single-row get\_video exists today). (= PER\_RALLY\_SELECTION P8 family; needs identity scoping.)", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 T8 + PER\_RALLY\_SELECTION.md §7 P8", "gated": "Owner (coord + identity-scoping gate)", "effort": "medium"}, {"step": "T9 ? S5 Settings screen: render config + capabilities; show the compute/storage target selector DISABLED ('single-box: local').", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 tracker T9 (depends T7, T8)", "gated": "Blocked (on T7, T8)", "effort": "small"}, {"step": "T10 ? Observability pass: correlation id (job\_id/upload\_id/request\_id) threaded end-to-end + structured logs at every new path + a thin metrics surface; logs inspected during CUJ replay (DoD item).", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 T10 + §5", "gated": "No", "effort": "medium"}, {"step": "T11 ? Retention sweep for output/ + uploads/ originals (currently never deleted) before any non-owner exposure.", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 tracker T11", "gated": "Owner (retention defaults §10 Q2: auto-delete window / keep proxies+intervals only)", "effort": "small"}, {"step": "T12 ? Ingest from bucket URI (s3://, gs://) ? engine half DONE (/api/process accepts URIs, M2/#280, verified); remaining is the SPA ingest call (= B-P2). NOTE: B-P2 SPA ingest was SHIPPED 2026-06-21 (engine #56 + DRY\_RUN\_CUJ §9), so T12 is effectively complete though the §9 tracker still shows ? ? flag for status update.", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 tracker T12 (?) vs DRY\_RUN\_CUJ.md §9 (B-P2 done)", "gated": "No (engine+SPA done; tracker row stale)", "effort": "small"}, {"step": "T13 ? Stand up a provider-agnostic GPU host + wire detector\_impl=ative (NativeWasbRunner); owner byte-parity gate (WSL CSV == native CSV) on a real GPU box.", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 tracker T13 (? owner; elevated to run in parallel) + §10 Q4", "gated": "Blocked (? owner ? owner provisions host+token; native WASB on engine branch feat/native-wasb-runner, trajectory\_hybrid still raises; PARKED-adjacent: needs a paid GPU machine/subscription)", "effort": "large"}, {"step": "T14 ? (reserved) Bucket-worker bridge: 'remote GPU as a setting' ? a net-new orchestration endpoint reflecting worker.infer outputs into the jobs/poll contract. (= PER\_RALLY\_SELECTION P9/P10-adjacent; reserved scale path.)", "sourceDoc": "LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 tracker T14 (? owner; depends T8, T12) + §10 Q5", "gated": "Blocked (? owner ? P5 bridge ownership undecided)", "effort": "large"}, {"step": "PER\_RALLY\_SELECTION P6 ? (engine) GET /api/health + offline banner; /api/compile returns download\_url (additive, low-risk; today returns a filepath only).", "sourceDoc": "PER\_RALLY\_SELECTION.md §7 P6 (not gated)", "gated": "No", "effort": "small"}, {"step": "PER\_RALLY\_SELECTION P7 ? (engine) GET /api/videos/{id}/rallies RESTful/cacheable alias returning the same {play\_segments} shape.", "sourceDoc": "PER\_RALLY\_SELECTION.md §7 P7 (not gated)", "gated": "No", "effort": "small"}, {"step": "PER\_RALLY\_SELECTION P9 ? (engine) source/clip stream (Range or /clip?start=&end=) + timeline + per-rally preview/thumbnail (no streamable source exists today, only the compiled reel).", "sourceDoc": "PER\_RALLY\_SELECTION.md §7 P9 (depends P8)", "gated": "Blocked (? owner; on P8)", "effort": "large"}, {"step": "PER\_RALLY\_SELECTION P10 ? (engine) PATCH /api/segments/{id} corpus correction loop (writes the dead candidate\_segments.human\_label/notes columns).", "sourceDoc": "PER\_RALLY\_SELECTION.md §7 P10 (depends P9)", "gated": "Blocked (? owner; on P9)", "effort": "medium"}, {"step": "REQUEST\_ACCESS\_FORM P2 ? finish Worker+D1 deploy (owner runs wrangler d1 create + deploy). NOTE: NEXT\_STEPS says the form is already LIVE on Cloudflare with D1 persistence + per-registration email ? so this row appears already-shipped; flag the tracker as stale.", "sourceDoc": "REQUEST\_ACCESS\_FORM.md §6 P2 (?) vs NEXT\_STEPS.md 'Where we are'", "gated": "Owner (deploy) ? likely already done", "effort": "small"}, {"step": "REQUEST\_ACCESS\_FORM P3 ? Anti-spam (Cloudflare Turnstile) + per-submission email notify (email-notify may already be live per NEXT\_STEPS).", "sourceDoc": "REQUEST\_ACCESS\_FORM.md §6 P3 (not gated)", "gated": "No", "effort": "small"}, {"step": "REQUEST\_ACCESS\_FORM P4 ? Tiny auth'd admin/export view to read D1 responses.", "sourceDoc": "REQUEST\_ACCESS\_FORM.md §6 P4 (owner-gated)", "gated": "Owner", "effort": "small"}, {"step": "Marketing-video cleanup: the real-UI CUJ

screencast (promo\_cuj.mp4/.gif + the bucket-CUJ screencast) shows real court footage + a dev 'P5' badge ? clean up before any public embed, and/or produce a polished cut.", "sourceDoc": "NEXT\_STEPS.md 'Marketing video on the site'", "gated": "Owner", "effort": "small"}], "ownerGated": [{"§10 Q1 edge-auth choice: Caddy basicauth (self-contained) vs cloudflared + Cloudflare Access (matches two-host design, no inbound ports) ? gates T2.", "§10 Q2 retention defaults: auto-delete originals after compile? keep proxies-intervals only? what window? ? gates T11.", "§10 Q3 per-video Gemini/GPU cost ceiling: what cap, hard-fail vs warn? (no cost ledger/ceiling exists today).", "§10 Q4 GPU host choice: D0 GPU droplet / GCE GPU VM / Cloud Run+GPU, and is a containerized worker worth building for Cloud Run (no Dockerfile exists; WASB's torch-1.11 conda env makes the image non-trivial)? ? gates T13. PARKED-adjacent: a real GPU machine needs a subscription.", "§10 Q5 P5-bridge ownership: who builds the bucket-worker orchestration endpoint and how it coordinates with the active engine worker sessions ? gates T14.", "DRY\_RUN\_CUJ option 3 (CPU-droplet CUJ confirmation): PARKED by the owner ? droplet un-provisioned, no GPU/subscription.", "PER\_RALLY\_SELECTION P8/P9/P10 (multi-video home + identity scoping, per-rally preview/timeline, corpus correction loop) are all ? owner-gated engine milestones.", "Nice-to-knows: console home (web/ vs engine static); review-playback streaming endpoint; per-user scoping timing; containerization vs runbook."], "staleOrInconsistent": [{"LOCAL\_APPLIANCE\_ARCHITECTURE.md §9 tracker T12 still shows ? ('engine done; SPA pending'), but the SPA half (B-P2 in-UI bucket ingest) was SHIPPED 2026-06-21 (web PR #56, recorded in DRY\_RUN\_CUJ.md §9 + git 4525c5d) ? T12 should be ?.", "LOCAL\_APPLIANCE\_ARCHITECTURE.md doc header says 'Status: DRAFT' and §9 T0 is ?, but pieces of the design (T12/B-P2, RallyPicker M0) have shipped and the dry-run CUJ has live e2e results ? the umbrella doc's DRAFT/? framing understates real progress on the slices it tracks.", "REQUEST\_ACCESS\_FORM.md §6 P2 is marked ? ('owner runs wrangler d1 create + deploy'), but NEXT\_STEPS.md 'Where we are' states the form is already LIVE on Cloudflare (Worker hidden-king-khelsutra + D1 khelsutra-access) persisting to D1 and emailing the owner per registration ? so P2 (and likely the per-submission email part of P3) reads as already-deployed; the REQUEST\_ACCESS\_FORM tracker is stale vs NEXT\_STEPS.", "LOCAL\_APPLIANCE\_ARCHITECTURE.md §3/§4.2 and the DoD reference GET /api/health as 'absent / net-new' in some places but the 2026-06-21 inline correction (§3) confirms it now EXISTS (engine M1/#268) ? partially reconciled but the §4.6 reuse map / §11 DoD still list /api/health under net-new, a residual inconsistency.", "PER\_RALLY\_SELECTION.md and LOCAL\_APPLIANCE\_ARCHITECTURE.md both carry a 'Created/Last updated: 2026-06-20' header while their bodies/changelogs include 2026-06-21 corrections and shipped work ? the Last-updated dates are stale."}], {"track": "localization+UX", "docsRead": ["C:/Users/avidu/Projects/khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/I18N\_PLAN.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/UI\_PROPOSAL.md"], "statusSummary": "Tracked in khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md §7 (source of truth). DONE (?): P0 (doc + pointers, #41/engine#275); B1 (SPA i18n scaffold + react-i18next + ICU + langdetector + switcher + persistence + jsdom/negotiation tests N1/N3-N7, #42); B2 (extract ~50 SPA literals -> en catalog + key-parity & no-bare-strings CI, #45); B4 (self-host OFL Noto Sans Kannada + Devanagari, #46); B5 (Kannada catalog, AI-draft, #47); B6 (Hindi catalog, AI-draft, #48). MILESTONE: the web/Studio SPA is fully trilingual en+kn+hi end-to-end ? a Kannada-default browser auto-renders Kannada; switcher walks kn->hi->en; proven by a Playwright screencast (locale kn-IN). Verified against the working tree: web/package.json carries i18next/react-i18next/i18next-icu/i18next-browser-languagedetector + @fontsource-variable noto-sans-kannada/devanagari; catalogs live at web/src/i18n/locales/{en, kn, hi}/common.json. 4 of 5 owner gates LOCKED/ADOPTED (D6 en+kn->hi, D7 OFL-font carve-out, D8 SPA-first, D9 catalogs co-located in khelsutra); only D-BACKEND-DEPTH (B7) remains open and is deferred. TODO/blocked (??): the entire site/ marketing surface (B3), recording-education kn/hi (A3, ? until B3), all 8 non-tech UX A-items except partial A9, native-speaker review of AI-drafted kn/hi, and engine P2 backend localization (B7, ?). NOT YET DONE per §9 DoD ? do not archive (both surfaces must be trilingual + UX baseline + manual font eyeball).", "nextSteps": [{"step": "B3 ? tiny shared t(key, vars) shim + data-i18n plumbing for site/ (en-only first), proving the shared catalogs work across both surfaces. Highest-priority remaining row: it unblocks A3 and is the only thing between the SPA milestone and a trilingual site/. Verified the site/ funnel pages (index.html, capture.html, own-model.html) are still English-only with no t()/data-i18n; the only localized site pages are the standalone hand-built academy print flyers flyer-kn/flyer-hi.html (from #44/#52), not the funnel.", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row B3 (depends B2, gate D-CATALOG-HOME satisfied)", "gated": "No", "effort": "medium"}, {"step": "A1 ? plain-language copy pass on the en catalog (de-jargon banners/footer; icon+text; non-color selection indicator). Unblocked now (depends only on B2, done).", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row A1; §4 UX-levers table (high impact / small effort)", "gated": "No", "effort": "small"}, {"step": "A2 ? localized WhatsApp share for the downloaded reel (SPA + site). Verified no wa.me/api.whatsapp wiring

exists in web/src or the site funnel today.", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row A2 (depends B1/B3); §4 (high/small, WhatsApp is THE channel)", "gated": "No", "effort": "small"}, {"step": "A4 ? mobile-first touch targets + make tooltip-text visible in the SPA (tiny targets + desktop layout + invisible tooltips on phones). No touch-target/min-h sizing markers found in web/src yet.", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row A4 (depends B1)", "gated": "No", "effort": "medium"}, {"step": "A5 ? friendly engine-error mapping over raw FastAPI detail (SPA client-side map), so CompileResult stops passing untranslated engine text. No errorMap/friendlyError helper present in web/src.", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row A5 (depends B2)", "gated": "No", "effort": "medium"}, {"step": "A8 ? background-processing reassurance + verdict/status as icon chips (SPA), honest expectations on slow uplinks.", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row A8 (depends B2)", "gated": "No", "effort": "small"}, {"step": "A9 ? tap-only filter controls alongside the English-grammar query bar (SPA). Partially advanced: a localized sample-query prompt (en/kn/hi) + an i18n-test rule for query parsing already shipped (#51), but the tap-only controls themselves remain. Note: parseQuery() is English-grammar-only; the §6.5 directive makes i18n tests a MERGE GATE the moment the parser gains semantic/Kannada/Hindi input.", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row A9 + §6.5 query-parsing directive (depends B2)", "gated": "No", "effort": "large"}, {"step": "A6 ? plain-language framing for the Request-Access GPU/storage questions (GPU/S3/GCS are intimidating). Site-side; effectively depends on B3.", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row A6 (depends B3)", "gated": "No", "effort": "small"}, {"step": "A7 ? fix English-source recording gaps (add a camera-placement diagram) before localizing, so the diagram is reusable across all 3 languages.", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row A7 (no dep)", "gated": "No", "effort": "medium"}, {"step": "A3 ? localize recording-education (capture page + homepage gist) to kn (then hi). Blocked on B3 (needs the site shim) plus B5; both D-LOCALES and D-FONT gates are satisfied so once B3 lands this is unblocked. capture.html verified fully English today.", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row A3 (depends B3/B5)", "gated": "Blocked (on B3 ? the site/ t() shim)", "effort": "medium"}, {"step": "Native-speaker review of the AI-drafted kn + hi catalogs (web/src/i18n/locales/{kn,hi}/common.json) before any wide public launch. Both catalogs are Gemini-drafted; a DONE/green CI proves key-parity+render, NOT translation quality (§6).", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §6 honest-limits + §7 B5/B6 notes + §9 DoD", "gated": "Owner (owner/native-speaker-paced review)", "effort": "small"}, {"step": "Owner manual cross-check on a real machine: set OS/browser to Kannada then Hindi and confirm BOTH surfaces render the right language AND the Noto fonts display (cloud CI can't render fonts; owner eyeball is the final font gate).", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §6.5 E2E + §9 DoD; I18N\_PLAN.md §6 Fonts", "gated": "Owner (owner-machine verification)", "effort": "small"}, {"step": "B7 (deferred, cross-repo) ? engine P2 backend customer-facing localization: backend/i18n/ with tr(lang,key,params), refactor validator messages to keys+params, localize HTTPException.detail + reporting/spec.yaml customer\_verdict/customer\_message, wire Accept-Language/?lang. Stays in the engine (badminton-highlight-indexer) repo; verified no backend/i18n exists there. Out of this project's DoD.", "sourceDoc": "FRIENDLY\_AND\_MULTILINGUAL.md §7 row B7 + D4/§8.3 D-BACKEND-DEPTH; I18N\_PLAN.md P2 + §4.4", "gated": "Owner (D-BACKEND-DEPTH open + cross-repo engine concern)", "effort": "large"}], "ownerGated": ["D-BACKEND-DEPTH (the only open decision in §8): how deep engine P2 backend localization goes for v1 ? recommendation is customer\_\* report text + validation-rejection reasons first, technical/operator text stays English, as a deferred phase-two engine-repo PR (B7). Not a v1 blocker.", "Native-speaker review of AI-drafted kn + hi catalogs is owner/native-speaker-paced (Gemini drafts are inference-only per the standing guardrail; quality is human-gated, not provable by CI).", "Final font-render eyeball (Kannada/Devanagari) must be done on the owner's real machine ? the cloud CI container can't render fonts.", "Counsel may still confirm the D-FONT OFL-for-media carve-out (owner LOCKED it as decision D7; B4 already shipped on that basis, so it is unblocked in practice)."], "staleOrInconsistent": ["§7 progress tracker in FRIENDLY\_AND\_MULTILINGUAL.md is mildly stale: its rows/changelog stop at PR #48-#49, but the khelsutra git log shows newer adjacent work merged ? #51 (localized sample-query prompt en/kn/hi + i18n query-parse test rule, partially advances row A9) and #52/#44 (localized academy print flyers flyer-kn.html/flyer-hi.html + nav link). Neither is reflected as a tracker row/status; the doc should note A9 is partially started and decide whether the standalone flyers count toward B3/A3.", "Catalog location vs. design: I18N\_PLAN.md §2.2 and FRIENDLY\_AND\_MULTILINGUAL.md §4 both diagram locales/ at repo root (D-CATALOG-HOME 'co-located in khelsutra'), but the shipped catalogs actually live under web/src/i18n/locales/{en,kn,hi}/common.json. They are NOT yet a shared top-level dir consumed by both surfaces ? so the cross-surface sharing premise of B3 (site/ reuses the SAME catalogs) is not yet realized; B3 will need to either hoist the catalogs or have the site shim import from web/src. Flag, don't fix.", "I18N\_PLAN.md still describes P1 as a 'static-UI

retrofit' of backend/api/static; that surface is retired (UI\_PROPOSAL B5; D3 drops it). The doc carries the retarget banner at the top, but the §3 phase table and §5 key-files still list backend/api/static as the P1 target ? internally consistent only via the banner." ]}, {"track": "rally-quality", "docsRead": ["C:/Users/avidu/Projects/badminton-highlight-indexer/docs/NEXT\_STEPS.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/RALLY\_QUALITY\_RESEARCH.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/HOW\_RALLY\_DETECTION\_WORKS.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/MULTI\_SIGNAL\_FUSION\_PLAN.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/EXPERIMENT\_HARNESS.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/GOLDEN\_REGRESSION\_FIXTURES.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/archives/roora-vendor/ROORA\_LABELING\_GUIDELINES.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/R2\_EVAL\_METRIC\_DESIGN.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/R1\_MILESTONE\_LAUNCH\_REPORT.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/QUALITY\_ITERATIONS.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/RALLY\_DETECTION\_GAP\_CLOSURE.md"], "statusSummary": "The rally-quality project (\"Bridge the local? Gemini rally-detection gap\", tracked in RALLY\_QUALITY\_RESEARCH.md §7) has its Tier-0 + Tier-1 windowing wins SHIPPED, and the active frontier has moved from \"fix over-merge\" to \"close the detection (precision/recall/rally-END) gap.\" \n\nDONE / shipped: E1 over-seg-aware leaderboard + frozen baseline (#185); W1 bounded windowing (max\_window\_duration cap + inactivity split, See et al.); W1.5 hard-cap (retained default-OFF); W2 net-crossings gate first-step (#188, do-no-harm, default-OFF); R2 task-faithful tolerance-match metric implemented as AUGMENT alongside tIoU (backend/eval/tolerance\_metrics.py + rally\_seg\_eval, #-2026-06-17); R5 blob-fallback for continuously-active clips (default-OFF stratum tool); P7 long-rally(>5s) eval lens + reel knob (default-OFF, measured n=15). The R1 MILESTONE flipped DEFAULT-ON (PR #204): W1 cap=6 + R4 rally-END inactivity trim (baseline? milestone R2 tolF1 +0.222, 12/0/1, p=0.0005 at n=13; net over-seg guard PASS), recorded in R1\_MILESTONE\_LAUNCH\_REPORT.md and independently reproduced (#214) + adversarially stress-tested. \n\nHarness fully built: experiment.py (Variant/compare/compare\_unlabeled/leaderboard), ablation.py, eval\_stats.py (bootstrap CI + paired sign-test), tolerance\_metrics.py (R2), rally\_seg\_eval.py one-command endpoint, regression.py gate. Golden Regression Fixtures (cross-machine CONTROL-only suite): P0 + M4 (quality-floor gate) + P3 (real Tier-0 commit) still ?. \n\nIn progress / next: corpus growth (n?6?15 done, target ?16; \"C\" row ?); the multi-feature (person/audio) learned scorer (W2 follow-on / P3) is the next measured step but is GATED on GPU-extracted golden features + .rallies.csv labels not present in this checkout. The current execution tracker for the remaining accuracy gap is RALLY\_DETECTION\_GAP\_CLOSURE.md §7 (P1-P6 all owner-gated/?), which re-prioritized the gap to DETECTION (precision #1, then recall, then rally-END), with Gate-A wiring named the cheapest precision win. \n\nKEY FINDING (n=2 golden, directional): LOCAL = high recall / low precision; GEMINI = high precision / low recall ? complementary ? fusion is the durable win. The gap is NOT boundary precision (R2/R3/R4 windowing does its job, |?|<1s on hits).\", \"nextSteps\": [{\"step\": \"Grow the golden corpus toward n?16 (multi-court badminton + ?3 matches/sport, table-tennis first); per video: label ? curate import-local --normalize --license Proprietary --fps <true fps> ? re-run Gen-0 harness. Single highest-leverage lever; gates the Tier-3 bets and the multi-feature scorer.\"}, {\"sourceDoc\": \"RALLY\_QUALITY\_RESEARCH.md §7 row C (? , ~15 now) + §6 'Throughout'; NEXT\_STEPS.md P0 #2\", \"gated\": \"Owner (owner labels golden clips; GPU/WSL + golden corpus are owner-side)\", \"effort\": \"large\"}, {\"step\": \"Wire rally\_gate Gate-A (net-crossings ?2) into the served production path as a config knob (the held-shuttle/service-feed structural fix) ? fusion.min\_crossings already exists default-0/OFF; promote only on measured lift. Named the cheapest, highest-confidence precision (G2) win.\"}, {\"sourceDoc\": \"RALLY\_DETECTION\_GAP\_CLOSURE.md §7 row P2 (? , owner-gated); MULTI\_SIGNAL\_FUSION\_PLAN.md §10 P2; RALLY\_QUALITY\_RESEARCH.md §7 W2\", \"gated\": \"Owner (and: served Gate-A measured neutral/negative on production windowing per litmus ? kept opt-in; needs golden re-measure to promote)\", \"effort\": \"small\"}, {\"step\": \"Extract person-velocity + shuttle\_player\_colocation person/audio features on the golden clips (default-OFF), then run the W2 multi-feature learned-scorer LOVO via experiment.compare + ablation.py to measure per-cue ?F1 (CI + sign-test) on the served windowing.\"}, {\"sourceDoc\": \"RALLY\_DETECTION\_GAP\_CLOSURE.md §7 P3; RALLY\_QUALITY\_RESEARCH.md §7 W2 ('multi-feature scorer next (GPU)'); MULTI\_SIGNAL\_FUSION\_PLAN.md §10 P3\", \"gated\": \"Blocked (on GPU-extracted golden features + .rallies.csv labels ? not present in this checkout; GPU/WSL owner-side)\", \"effort\": \"medium\"}, {\"step\": \"Run the R2-vs-tIoU ablation experiment on the golden corpus to promote R2 tolF1 from AUGMENT to the official promotion GATE (show R2 ranks candidates at least as sensibly as tIoU before swapping); then deprecate-with-docs (never delete) tIoU as a secondary trend-line. Also surface the tolerance\_sweep helper as a rally\_seg\_eval CLI flag and add the dual-set R2 aggregate + over-seg guard to run\_regression.py.\"}, {\"sourceDoc\": \"R2\_EVAL\_METRIC\_DESIGN.md §2.3.5/§5/§6/§7; NEXT\_STEPS.md P1 #8\", \"gated\": \"Owner (gate-swap is owner decision) + needs GPU-extracted golden features for the

full corpus ablation", "effort": "medium"}, {"step": "IAA (inter-annotator agreement) study: a 2nd labeler re-labels 273 already-golden videos; set ?\_start/?\_end to ~the 90th-percentile pairwise human |?start|/?end| (provisional ?\_start=2.0/?\_end=1.5 frozen until then). Prerequisite for trusting R2's ? choice.", "sourceDoc": "R2\_EVAL\_METRIC\_DESIGN.md §2.4 + §6 (ADOPTED 2026-06-17)", "gated": "Owner (needs a 2nd labeler / Roora A/A overlap)", "effort": "medium"}, {"step": "Lever A ? player-state rally-END rule (the R4 successor for G3 quick-return-after-point): generalize R4 from shuttle-only to shuttle OR bilateral player motion + hard-terminate on a shuttle-picked-up / sustained-Visibility=0 anchor with reverse-timeline search; measure ?F1 / ?merge-rate / ?|?end| vs R4.", "sourceDoc": "RALLY\_DETECTION\_GAP\_CLOSURE.md §4 Lever A + §7 P4 (?, owner-gated); RALLY\_QUALITY\_RESEARCH.md §6 M1", "gated": "Blocked (depends on P3 person features on golden clips) + Owner", "effort": "medium"}, {"step": "Lever B/C ? recall + complementary (local+Gemini / shuttle+person) fusion: measure recall lift with no precision regression; pick the next videos to label by max local-vs-Gemini disagreement (active learning).", "sourceDoc": "RALLY\_DETECTION\_GAP\_CLOSURE.md §4 Lever B/C + §7 P5; RALLY\_QUALITY\_RESEARCH.md §4.3/§4.4", "gated": "Blocked (depends on P3) + Owner", "effort": "medium"}, {"step": "M2 ? per-venue threshold calibration (velocity\_threshold + merge\_gap) against golden, targeting the heuristic's multi-court fragility (0.157) toward Gen-0's 0.561 without harming clean-court.", "sourceDoc": "RALLY\_QUALITY\_RESEARCH.md §6 Tier-2 M2 + §7 row M2 (?)", "gated": "No (actionable once golden multi-court features available; logic is CPU)", "effort": "medium"}, {"step": "W3 ? record the standing decision to keep the scorer parameter-light (numpy LogisticRegression; no learned temporal net until corpus ? now) per TemporalMaxer evidence. Documentation-only.", "sourceDoc": "RALLY\_QUALITY\_RESEARCH.md §6 Tier-1 W3 + §7 row W3 (?); §5.1 TemporalMaxer", "gated": "No (actionable now)", "effort": "small"}, {"step": "Golden Regression Fixtures: finish M4 (quality-floor gate + first regression\_baseline.json with windowing fingerprint) and P0 (leak remediation ? scrub real player/venue names + personal paths in eval\_baselines/heuristic\_lovo\_n6.json, gemini\_wrapper\_2026-06-10.json, backend/eval/eval\_partial\_labels.py + add required CI leak-scan), then optionally P3 real Tier-0 commit.", "sourceDoc": "GOLDEN\_REGRESSION\_FIXTURES.md §7 rows P0/M4/P3 (?) + §0 'two findings already actionable'", "gated": "P0/M4 = No (owned-data hygiene, actionable now); P3 = Owner (per-video clearance list)", "effort": "medium"}, {"step": "Tier-3 research bets (corpus-gated stretch): R1 owned segmenter with an explicit boundary head (port ASRF/TriDet idea, TRAINED not grafted ? refuted as a graft) ? pilot only at corpus ~216?30 matches after W1/W2 set the heuristic ceiling; R2 timestamp/sparse-supervision label-efficiency pilot. Default DoD completes at Tier 0?2; Tier-3 spins into the owned-model milestone.", "sourceDoc": "RALLY\_QUALITY\_RESEARCH.md §6 Tier-3 + §7 rows R1/R2 + §7.4 Tier-3 scope decision", "gated": "Blocked (corpus ?16?30) + Owner", "effort": "large"}], "ownerGated": ["Set the ship-gate / success-bar target (NEXT\_STEPS #172): proposed shape 'LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with paired-sign p<0.05 vs heuristic', or per RALLY\_QUALITY\_RESEARCH §7.1 'close ?50% of the heuristic?Gemini multi-court gap (local multi-court F1 ? ~0.41)'. Needs corpus growth; owner to ratify/adjust.", "Grow the golden corpus (owner labels clips; golden corpus + GPU/WSL are owner-side) ? this gates the multi-feature scorer, M2 calibration, the R2 ablation gate-swap, and all Tier-3 bets.", "Approve starting RALLY\_DETECTION\_GAP\_CLOSURE.md P1-P6 (Gate-A wiring, person-feature extraction, Lever A/B/C) ? all rows marked ? owner; one PR per row, default-OFF, promote only on measured golden lift.", "Court-mask source decision for M1 serve-start classifier + Gate-B / court-localization (manual per-camera polygon vs auto homography) ? MULTI\_SIGNAL\_FUSION\_PLAN §9.1, owner-gated.", "Decide which lever first once labels land (RALLY\_DETECTION\_GAP\_CLOSURE §8): no-model wins (Gate-A wiring + shuttle-invisibility end-anchor) vs straight to person-velocity; and the per-video target (at-par-with-Gemini-on-every-clip vs an absolute F1 floor).", "Re-confirm the GOLDEN\_REGRESSION\_FIXTURES Tier-0 risk acceptance on the CORRECTED basis (the shipped freeze commits the full de-attributed per-frame trajectory.csv ? a de Montjoye fingerprint ? NOT ~5 scalars; the 2026-06-16 acceptance was recorded against the inaccurate '5 scalars' basis) before any REAL Tier-0 fixture (P3) is committed.", "Approve the eventual R2-as-gate swap + tIoU deprecation-with-docs after the validating ablation (R2\_EVAL\_METRIC\_DESIGN §7)."], "staleOrInconsistent": ["RALLY\_QUALITY\_RESEARCH.md §7 progress tracker is partially STALE vs the rest of the track: its §7.1 progress table and §7.2 checklist still say golden 'n=6 aggregate' and list W1.5/W2 as ? with 'multi-feature scorer next (GPU)', while QUALITY\_ITERATIONS.md, R1\_MILESTONE\_LAUNCH\_REPORT.md and GOLDEN\_REGRESSION\_FIXTURES.md all state the corpus has grown to n=15 and that R1 milestone (W1 cap6 + R4) shipped DEFAULT-ON (PR #204), R2 metric is implemented, and R5/P7 shipped ? none of which are reflected as ? rows in §7. The §7 table should be reconciled (add R2/R3/R4/R5/P7 rows + flip the C-row n and re-baseline) or explicitly point to RALLY\_DETECTION\_GAP\_CLOSURE.md §7 as the now-live execution tracker.", "NEXT\_STEPS.md doc header self-identifies the rally-quality detail section as the '2026-06-13 plan; still valid' and its 'Next-session/GPU-box pickup' block is dated 2026-06-15"]

(pre-R1/R2/R5/P7) ? it predates the shipped milestone and reads as if W1/W2 are the immediate frontier, whereas the live frontier is RALLY\_DETECTION\_GAP\_CLOSURE.md (2026-06-18). The top 'Where we are' header was refreshed 2026-06-21 but the rally-quality detail body was not.", "Golden corpus count is cited inconsistently across docs as a moving target: NEXT\_STEPS 'n=6' (stale), QUALITY\_ITERATIONS/EXPERIMENT\_HARNESS '15', R1 launch report 'n=13', RALLY\_DETECTION\_GAP\_CLOSURE 'n=15' corpus (n=2 golden ground-truth scored)". The distinction (labelled manifest size vs GPU-feature-extracted vs golden-ground-truth-scored) is real but the bare 'n' is easy to misread; worth a single canonical count statement.", "RALLY\_QUALITY\_RESEARCH.md §7.1 still carries a 'TODO: compute exact % gap-to-Gemini closed needs Gemini measured on the same golden set + metric' ? and RALLY\_DETECTION\_GAP\_CLOSURE §3 notes Gemini intervals for the Boxhill clips live only in the owner-side scorer DB (not committed), so the headline gap-closure % the project's DoD requires is still uncomputed.", "GX010141 golden numbers in RALLY\_DETECTION\_GAP\_CLOSURE §3 are flagged as labeled BEFORE ~3 rally-annotator-tool bugs were fixed (GX010137 is post-fix) and 'may shift on a re-export' ? a self-flagged data caveat that should be re-verified before any hard conclusion." ]], {"track": "owned-model+strategy", "docsRead": ["C:/Users/avidu/Projects/badminton-highlight-indexer/docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/COMMERCIALIZATION.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/archives/field-pmf/PMF\_MARKET\_RESEARCH.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/archives/field-pmf/PMF\_FIELD\_SIGNALS.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/DEFERRED\_HARDENING\_PLAN.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/ROADMAP.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/BACKLOG.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/NEXT\_STEPS.md"], "statusSummary": "Owned-model track is at Phase 0 (Foundation), partially executed and partially still owner-gated. DONE: M0.2 corpus grown to n=6 (262 rallies, Proprietary, in collector); the Gen-0 TAL harness (training/gen0/harness.py) ships and produced artifact v0.1.0 at LOVO F1 0.626 (passed the 0.480 heuristic floor but sign test 5/6 wins p=0.109 -> not significant, ship=False); train/eval split guard core (backend/eval/splits.py, #171); ship-gate provisional blocker (#170); doc-consistency sweep (#174); P1/P2 of the rally-quality fusion track (PersonDetector reuse + FusionSegmenter + Gate A signal, registry fusion\_hybrid, default-OFF). The Gemini quality battery is complete and now drives strategy (multi-court Gemini wrapper = 0.66 = the owned model's ship target; clean footage 0.93 is conceded to Gemini). NOT STARTED / still gated (per NEXT\_STEPS \"Leaving bulky/M2\"): M0.1 event-index DB migration (verified: DB at user\_version=6, no highlights table / rally\_ordinal / fault\_tags / ending\_reason yet); M0.3 Gemini-silver purge (backend/eval/distill\_local.py still present); M0.4 ActionFormer/TemporalMaxer scorer swap; the real P3 person-fusion LOVO measurement (needs GPU-extracted golden features). STRATEGY track: licensing posture ratified (MIT/Apache/BSD; Qwen3-VL primary base, Gemma-4 AMBER/bench-only, Gemma 3 rejected); DEFERRED\_HARDENING\_PLAN fully completed (#16 async jobs DONE PR #87, #7 packaging DONE PR #167, #13 train/eval split moved to BACKLOG with core landed); PMF beachhead (India badminton academies) defined with a pre-registered field-validation decision framework but UNVALIDATED (owner-side field work). Sequencing discipline still holds: no owned-model implementation beyond greenlit M0.x without explicit owner go.", "nextSteps": [{"step": "Grow the golden corpus: multi-court badminton (target distribution + where the 0.66 ship target lives) + >=3 matches per new sport (table tennis first; WASB transfers at F1 0.909). Adult sessions only for the training pool. The single highest-leverage move; every other lever feeds on it. At n=10 with 9 wins the paired sign test reaches p=0.011 (significance reachable).", "sourceDoc": "NEXT\_STEPS.md Prioritized pending P0 #2 + owned-model detail #1", "gated": "Owner (owner-side data collection + consent)", "effort": "large"}, {"step": "Full-corpus training (n>=5) on the GPU box -> a real weights/<ver>/ bundle (the 0.1.0-l4 bundle is n=2 plumbing-only).", "sourceDoc": "NEXT\_STEPS.md Prioritized pending P0 #3", "gated": "Owner (GPU box + corpus) / Claude can drive", "effort": "medium"}, {"step": "Run the real P3 person-fusion LOVO measurement: backend.eval.fusion\_golden to extract \*\_golden\_features.json for the 6 golden videos (GPU box), then backend.eval.fusion\_compare + backend.eval.ablation --granularity group for real person-feature delta-F1 + bootstrap CI + paired sign test; copy JSONs to output/ to satisfy the skipif litmus (tests/test\_ablation.py).", "sourceDoc": "NEXT\_STEPS.md GPU-box pickup 2026-06-15 #1-3", "gated": "Blocked (on GPU-extracted golden features + .rallies.csv labels, owner-side only)", "effort": "medium"}, {"step": "Set the ship-gate F1@tIoU target for 'very high P/R' (floor = beat heuristic LOVO 0.480 necessary-not-sufficient; multi-court reference = Gemini wrapper 0.66; clean ceiling 0.93 conceded). Suggested shape: LOVO F1@tIoU0.5 >= 0.70 on multi-court folds with paired-sign p<0.05 vs heuristic. (Issue #172.)", "sourceDoc": "OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md M0.4 + Decisions table #1 + NEXT\_STEPS #2/#7", "gated": "Owner (needs corpus growth to inform)", "effort": "small"}, {"step": "M0.4 scorer swap: replace the Gen-0 LogReg with a lightweight temporal head inside the existing harness + per-video threshold calibration. Prefer

TemporalMaxer (parameter-free max-pool, ~2.8x fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review. \$0/local on the 2070S.", "sourceDoc": "OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md M0.4 + NEXT\_STEPS owned-model #3 + P2 #10", "gated": "Owner (greenlight item; bulky/M2 - no start without owner go)", "effort": "medium"}, {"step": "M0.3 compliance cleanup (hard blocker before any training run): purge the Gemini-silver training path in backend/eval/distill\_local.py (trains a classifier on Gemini-silver CSV -> output/local\_rally\_model.json) + threshold distillation in calibrate\_local.py; retarget to human + open-teacher (own-model/WASB) labels. Verified distill\_local.py still present.", "sourceDoc": "OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md M0.3(i)", "gated": "No (actionable; greenlight item, but Claude-doable code work) / sequence before any training run", "effort": "small"}, {"step": "M0.1 event-index DB migration (the conversational-QA seam): add a PRAGMA user\_version migration extending play\_segments + a new highlights table with the per-rally event-index schema {rally\_ordinal, sport, start, end, shot\_count, ending\_reason/fault\_tags, derived\_clip\_ref, provenance, confidence}; reserve the fault/event taxonomy now. Verified NOT done: DB at user\_version=6, no highlights table or these columns.", "sourceDoc": "OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md M0.1 + conversational-QA seam section", "gated": "Owner (greenlight item; bulky/M2 - no start without owner go)", "effort": "medium"}, {"step": "Quantify cost-to-Gen-2 (the largest open risk): benchmark real VRAM/token/\$ for VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 6-8 GPUs) BEFORE standing up any owned serving stack. Also the \$/call crossover (self-hosted VLM vs Gemini).", "sourceDoc": "OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md Phase 2 pre-serving gates + BACKLOG owned-model", "gated": "Owner (gates Phase-2 commitment; budget unquantified)", "effort": "medium"}, {"step": "Finish the train/eval separation guard wiring: backend/eval/splits.py core + tests landed (#171) but STILL DEFERRED is wiring the guard into the training entrypoint (distill\_local/training) and run\_regression.py (baseline strictly on eval split), plus adding the collector curate export manifest 'split' field (cross-repo, ADR-002) and the leaked-manifest integration test.", "sourceDoc": "BACKLOG.md Golden-set subproject 'Train/eval separation guardrails' (was DEFERRED\_HARDENING #13)", "gated": "Owner (needs a collector-side manifest change + dedicated design session)", "effort": "medium"}, {"step": "Multisport: ingest Rooro pickleball/TT CSVs (curate import-local --normalize); merged-prototype LOVO across badminton+TT once TT has >=3 matches; identify a clean MIT/Apache/BSD pickleball ball detector (none identified yet). TT-on-WASB serving stays gated by C3 provenance (multiplies per sport).", "sourceDoc": "OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md Multisport + NEXT\_STEPS owned-model #4 / P2 #11", "gated": "Blocked (TT needs >=3 matches; pickleball blocked on a clean ball detector; serving gated by WASB C3)", "effort": "large"}, {"step": "WASB C3 weight-provenance resolution (commercial-ship blocker, multiplies per sport): resolve via own-trained shuttle detector OR a clean MIT/Apache/BSD detector swap. Note own-weights is an un-scoped NEW workstream (ADR-002: our temporal labels can't train the detector - needs per-frame coordinate labels), not delivered by ROADMAP Phase 4. Gates SHIP/sharing, not Gen-0 research.", "sourceDoc": "OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md M0.3(ii) + COMMERCIALIZATION C3 + NEXT\_STEPS P2 #13", "gated": "Owner + counsel (owner said yes to making it a first-class action item; resolution path un-scoped)", "effort": "large"}, {"step": "Rename the repo off 'badminton' (project is multisport; owner-emphatic ASAP). Touches remote name + README/CLAUDE.md framing + hardcoded URLs.", "sourceDoc": "OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md Decisions #5 + BACKLOG + NEXT\_STEPS P2 #14", "gated": "Owner (owner action, before broad external sharing)", "effort": "small"}, {"step": "PMF field validation: run the C13 academy discovery campaign (n=15 owner/head-coach sheets, pre-registered PROCEED/PIVOT/FALSIFIED thresholds) using the completed field kit; then a camera pilot at the 2-3 warmest academies + a fake-door/paid-pilot WTP test (B2C ~\$99/yr, B2B academy ~\$600/yr) before any money is spent on build.", "sourceDoc": "PMF\_FIELD\_SIGNALS.md s3 decision rules + PMF\_MARKET\_RESEARCH.md s6 + COMMERCIALIZATION C13", "gated": "Owner (business/field track, owner-side)", "effort": "large"}, {"step": "Codec output economics (C5, P0 commercial): switch highlight output from H.264+AAC (libx264 in backend/pipeline/stitcher/compiler.py) to royalty-free AV1+Opus; requires provisioning an Ada-class GPU (L4/L40) for HW AV1 encode (Ampere/Turing must use SVT-AV1 CPU).", "sourceDoc": "COMMERCIALIZATION.md s4 + dashboard C5", "gated": "Owner (decision: provision Ada-class GPU?)", "effort": "medium"}], "ownerGated": ["Greenlight the next Phase-0 milestone(s): plan recommends M0.1 (corpus + event-index DB migration) then M0.4 (TAL harness scorer swap). No owned-model implementation beyond already-greenlit M0.x starts without explicit owner go (NEXT\_STEPS explicitly lists M0.4/M0.1/M0.3/multisport/repo-rename under 'Leaving - no start without owner go').", "Separate explicit training-data opt-in + exclude-minors + per-clip consent ledger before any training that uses user data (guardrail; adult-only sessions for the training pool).", "Set the ship-gate F1@IoU target for 'very high P/R' (#172) - currently undefined; needs corpus growth to inform.", "Confirm the collector as the corpus + event-store system-of-record (Decisions table #4 still 'owner: confirm?').", "Accept the

conversational-QA recommendation: build the event-index seam now, punt the NL-parser/model (Decisions #6 'rec accepted?').", "Risk-acceptance for using an open VLM base (Qwen3-VL) whose pretraining provenance is unauditale - our no-paid-LLM rule is enforceable only at our fine-tuning layer, not the base's pretraining.", "Budget sign-off / quantification for cost-to-Gen-2 (largest open risk, currently unquantified) before standing up any owned serving stack.", "WASB C3 weight-provenance + per-checkpoint/codec-royalty (AVC/HEVC) legal sign-off (counsel) before commercial launch/sharing; provision an Ada-class GPU for the AV1 codec switch.", "PMF beachhead is unvalidated - owner-side field campaign + fake-door pricing must run before build money is spent.", "Rename repo off 'badminton' (owner action).", "staleOrInconsistent":["ROADMAP.md still presents Phase 4 as the candidate-window classifier narrative and lists Phase 3 'in progress' / Phase 4 'Next'; this is superseded by OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md (TAL head) - ROADMAP carries a top warning banner pointing to the plan, but its body/status table was not updated to reflect that the Gen-0 TAL harness already ships (LOVO 0.626). Narrative-vs-current-state drift.", "COMMERCIALIZATION.md header says 'Last refreshed: 2026-06-07' and its dashboard/See-also still link DEFERRED\_HARDENING\_PLAN.md async (C12) and packaging (C9) as 'pending approval', but those are now COMPLETED (async #87, packaging #167). C9 row text is updated to reflect packaging done, but the See-also line and C12 framing lag. Stale refresh date vs landed work.", "COMMERCIALIZATION.md (and OWNED\_MODEL plan M0.4 / BACKLOG) reference paths that have since moved or evolved: COMMERCIALIZATION links 'PMF\_MARKET\_RESEARCH.md' / 'I18N\_PLAN.md' / 'GOLDEN\_SET\_VENDORING.md' as if at docs/ top level, but they now live under docs/archives/field-pmf/, docs/, and docs/archives/roora-vendor/ respectively (link rot from the doc-hygiene archival moves tracked in DOC\_STATUS s4).", "OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md M0.4 still frames the scorer as ActionFormer/ASFormer; the 2026-06-17 external review (captured only in NEXT\_STEPS + RALLY\_QUALITY\_RESEARCH s5.6) now prefers TemporalMaxer. The authoritative plan doc has not been updated to reflect that preference.", "BACKLOG.md header 'Last updated: 2026-06-10' and its ship-gate note ('learned is now 0.626 post metric-fix') predate the Gen-0 harness artifact v0.1.0 record and the full Gemini battery; BACKLOG was not refreshed alongside NEXT\_STEPS (2026-06-21). The DEFERRED\_HARDENING\_PLAN async '(pending approval)' phrasing in COMMERCIALIZATION See-also is the clearest contradiction with the now-COMPLETED stub.]], {"track": "cross-cutting", "docsRead": ["C:/Users/avidu/Projects/badminton-highlight-indexer/docs/DOC\_STATUS.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/NEXT\_STEPS.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/README.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/CODE\_MAP.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/COMPUTE\_DECOUPLED\_SERVING/README.md (via origin/master)", "C:/Users/avidu/Projects/khelsutra/docs/README.md", "C:/Users/avidu/Projects/khelsutra/docs/DRY\_RUN\_CUJ.md (via git grep)", "C:/Users/avidu/Projects/khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md (via git grep)"], "statusSummary": "Two repos, several active tracks; the cross-cutting register that holds it together is engine docs/DOC\_STATUS.md (LIVING, last full review 2026-06-21) ? but origin/master has moved AHEAD of it since that review. ACTIVE / owner-gated tracks per DOC\_STATUS: (1) COMPUTE\_DECOUPLED\_SERVING (the headline track ? one pure DETECT?WINDOW?STITCH core + ComputeBackend/StorageBackend registries + truly-local/cloud-serving/cpu-mock profiles; ADR-014 ratified, M0 approved). DOC\_STATUS credits only 'M1?M4 SHIPPED + M5+ batch authorized', but origin/master in fact has M1?M6 + M8 merged (#279/#280/#282/#283/#286/#287/#288) plus bucket-ingest B-P1/B-P2 fixes (#284,#289); the project's own \$7 tracker (the source of truth) is current ? M5/M6/M8 = ? (code shipped, real-cloud/owner residuals pending), M7/M9/M10/M11/M12 = ?. (2) Rally-quality / owned-model R-series cluster (RALLY\_QUALITY\_RESEARCH foundation + R2 metric + RALLY\_DETECTION\_GAP\_CLOSURE + PROMINENT\_COURT\_DETECTION + MULTI\_SIGNAL\_FUSION P1 done + EXPERIMENT\_HARNESS P1 + QUALITY\_ITERATIONS) ? ACTIVE, owner-gated, blocked on growing the golden corpus + GPU-extracted golden features; do NOT archive until R2 promoted-or-rejected as the gate and \$7.1's single quality number is attributed. (3) PER\_VIDEO\_WORKSPACE (P0 #264 done, P3 base de-hardcode done via #282; P1 v6 migration must be co-authored with M8's v6 ledger). (4) EXECUTION\_POLICY (P1 shipped, P3b next; reused by Cloud Run dispatch). (5) Owner-gated design-only: OWNED\_MODEL\_IMPLEMENTATION\_PLAN (authoritative roadmap), PERSONALIZATION\_PLAN (Phase-0), DESKTOP\_APP\_PLAN (nothing built), GOLDEN\_REGRESSION\_FIXTURES, ANALYZERS\_AND\_RECONCILERS, PROMINENT\_COURT\_DETECTION. Cross-repo: the UX + localization track lives in khelsutra (FRIENDLY\_AND\_MULTILINGUAL = engine I18N\_PLAN + UI\_PROPOSAL UX-half; DRY\_RUN\_CUJ = bucket?reel CUJ; PER\_RALLY\_SELECTION M0 complete; LOCAL\_APPLIANCE\_ARCHITECTURE umbrella). On the khelsutra side the SPA is fully trilingual (en+kn+hi, milestone 2026-06-21) and the DRY\_RUN\_CUJ both DoD boxes are ? (Path A + Path B live e2e done, #57) ? but khelsutra docs/README.md still bills DRY\_RUN\_CUJ as the open '? active goal'. Engine local checkout is on feat/gcp-ops-agent-and-dryrun-guide (Ops-Agent codification, NEXT\_STEPS P1 #6), 1 ahead / 0 behind origin/master.", "nextSteps":["{"step": "Refresh DOC\_STATUS \$3a + \$4 (and NEXT\_STEPS) to reflect that COMPUTE\_DECOUPLED\_SERVING M5/M6/M8 + bucket-ingest B-P1/B-P2 are merged on origin/master ? the register's 'M1?M4 SHIPPED / M5+ authorized' line is stale (last review 2026-06-21 predates

#286/#287/#288/#289).", "sourceDoc": "engine docs/DOC\_STATUS.md §3a row 33 + §4 #6; NEXT\_STEPS P0 #1 / P1 #4", "gated": "No", "effort": "small"}, {"step": "Serving track ? drive the ? rows to ?: M5 truly-local runlog, M7 cloud-serving e2e on L4 + DEPLOY\_REPORT, then M9 edge-auth, M11 flywheel, M12 gdrive-api OAuth backend (one PR per milestone, default-OFF, off origin/master).", "sourceDoc": "engine COMPUTE\_DECOUPLED\_SERVING/README.md §7 (M5?M12)", "gated": "Owner", "effort": "large"}, {"step": "Codify the GCP Ops Agent into deploy/gcp so every VM auto-enables observability (this is the work on the current local branch feat/gcp-ops-agent-and-dryrun-guide ? finish + PR it).", "sourceDoc": "engine NEXT\_STEPS P1 #6", "gated": "No", "effort": "small"}, {"step": "Per-video workspace P1?P6 ? author the v6 DB migration ONCE jointly with M8's v6 cost-ledger migration; route detector/ + tmp/ per video (output/chunks\_\* base de-hardcode already done via #282).", "sourceDoc": "engine NEXT\_STEPS P1 #5; DOC\_STATUS §3a PER\_VIDEO\_WORKSPACE", "gated": "No", "effort": "medium"}, {"step": "Grow the golden corpus (multi-court badminton + >=3 matches/sport, TT first) ? the single highest-leverage move that unblocks both the rally-quality R-series and full-corpus owned-model training.", "sourceDoc": "engine NEXT\_STEPS P0 #2", "gated": "Owner", "effort": "large"}, {"step": "R-series: extract GPU golden features (.rallies.csv labels), then promote-or-reject R2 to F1 as the regression gate (the ablation) and attribute RALLY\_QUALITY\_RESEARCH §7.1's single quality number ? only then archive the R-cluster.", "sourceDoc": "engine NEXT\_STEPS P1 #8 / P2 #9; DOC\_STATUS §3a R2 + §4 #3", "gated": "Blocked (on GPU-extracted golden features + corpus growth)", "effort": "medium"}, {"step": "khelsutra FRIENDLY\_AND\_MULTILINGUAL: take SPA trilingual win to the site/ surface (B3 shared t() shim + D-CATALOG-HOME), then the A-series non-tech UX items + native-speaker review of the AI-drafted kn/hi catalogs before public launch (don't archive until BOTH surfaces are trilingual).", "sourceDoc": "khelsutra docs/FRIENDLY\_AND\_MULTILINGUAL.md §7 (B3 + A-series) + §9 DoD", "gated": "No", "effort": "medium"}, {"ownerGated": ["COMPUTE\_DECOUPLED\_SERVING owner/counsel residuals (NOT blocking Claude's default-OFF code): counsel-review ToS/DPA (D10/M9, M11), real GCP prices + FX for the M8 pricebook calibration, the Google OAuth app for M12, Cloudflare Access team-domain/AUD + Cloud Run ingress-lock (M9); real cloud verification authorized only within the D17 \$100/Rs10k cap (confirm cmd+hourly cost before each spend, DELETE every resource after).", "Set the #172 ship-gate target (e.g. LOVO F1@tIoU0.5 >= 0.70 on multi-court folds with paired-sign p<0.05) ? needs corpus growth first.", "Grow the golden corpus (owner-side: golden labels, multi-court footage, adults-only training pool).", "OWNED\_MODEL\_IMPLEMENTATION\_PLAN milestones beyond greenlit M0.x; M0.4 scorer swap (prefer TemporalMaxer); multisport (TT first); DESKTOP\_APP\_PLAN P0 spike; PROMINENT\_COURT\_DETECTION / ANALYZERS\_AND\_RECONCILERS / GOLDEN\_REGRESSION\_FIXTURES designs.", "WASB-C3 weight provenance (legal ? gates SHARING a trained model); rename repo off 'badminton'; retire the shared D0 droplet + D0 token.", "khelsutra B7 engine P2 backend customer-facing localization (D-BACKEND-DEPTH, owner-gated).", "staleOrInconsistent": ["engine docs/DOC\_STATUS.md §3a (row 33) + §4 #6 say COMPUTE\_DECOUPLED\_SERVING is 'M1?M4 SHIPPED (#279/#280/#282/#283); M5+ default-OFF code batch authorized' ? but origin/master has since merged M5 (#286), M6 (#287), M8 (#288), the B-P1 guard (#284) and the bucket-ingest compile fix (#289). The register's last full review (2026-06-21) predates these. The project's own §7 tracker IS current (M5/M6/M8=?, M7/M9-M12=?); DOC\_STATUS + NEXT\_STEPS lag it.", "engine docs/NEXT\_STEPS.md P0 #1 / P1 #4 still describe the M5?M12 batch as upcoming ('M5 truly-local -> M6 ... -> M8 ledger') whereas M5/M6/M8 code is already merged ? the prioritized list needs re-baselining to M7/M9/M11/M12 + the real-cloud residuals.", "khelsutra docs/README.md still bills DRY\_RUN\_CUJ.md as the open '? active goal (owner directive 2026-06-21)', but that doc's DoD §6 is fully ? (Path A + Path B live in-UI bucket?reel e2e DONE, #57) ? README should mark it delivered.", "khelsutra docs/README.md does NOT index docs/ANNOTATE\_AND\_TRAIN.md (Status DRAFT, owner-gated on scope, 2026-06-20) ? an untracked doc missing from the index (PORTABILITY.md is indexed; ANNOTATE\_AND\_TRAIN is not). Minor: this gap means the khelsutra-side index is not a complete register.", "Cross-repo consistency: engine DOC\_STATUS §3a row 49 marks I18N\_PLAN 'in implementation (adapted)' and §4 #5 marks the UI\_PROPOSAL/I18N pointer rows resolved ? accurate in direction, but the engine register does not reflect that the khelsutra SPA already hit 'fully trilingual' (2026-06-21 milestone); the engine side reads more tentative than the khelsutra reality."]]]

Produce: a headline; exactly what's next on the track I've been on (in order); ONE prioritized cross-track list (actionable-now first, each tagged track + gate + why); the owner decisions that unblock the most; a short serving-track pointer; and consolidated stale-doc flags. Be concrete and cite source docs. Return structured JSON.

## 2. assistant (2026-06-21T13:58:50.104Z)

I have all the per-track scout data I need to synthesize the roadmap. The task is pure synthesis

from the provided scouts plus the session context (appliance/CUJ + localization track recently driven; droplet PARKED; serving scanned separately). Let me produce the structured output.

### 3. user (2026-06-21T14:00:28.285Z)

Structured output provided successfully

### 4. assistant (2026-06-21T14:00:47.250Z)

I've completed the synthesis and returned the prioritized roadmap via StructuredOutput. The key findings:

**\*\*Headline:\*\*** The dry-run CUJ track I've been driving is essentially done end-to-end (Path A + Path B/B-P2 live with screencast, SPA fully trilingual en+kn+hi), the only remaining CUJ box (CPU droplet, option 3) is owner-PARKED, so the sensible pivot is onto the active "Friendly & Multilingual" directive ? starting with B3 (shared site/ t() shim) and A1 (plain-language en copy pass).

**\*\*My track, in order:\*\*** B3 ? A1 ? A2 ? A4 ? A5 ? A8 ? A9 ? A7 ? A3/A6, then the two owner gates (native-speaker kn/hi review + owner-machine font eyeball). Critical detail flagged: catalogs live at `web/src/i18n/locales/{en,kn,hi}/common.json`, not the shared top-level dir the design diagrams assume, so B3 must hoist or import them.

**\*\*Cross-track prioritization\*\*** leads with the ungated localization/UX rows, then cheap doc-hygiene refreshes, then the ungated engine code work (M0.3 Gemini-silver purge, per-video v6 migration co-authored with M8, W3 doc, Golden-fixtures P0/M4), with all GPU/corpus/owner-gated rally-quality and owned-model work below.

**\*\*Owner decisions that unblock the most:\*\*** confirm localization stays the sole focus; native-speaker kn/hi review; corpus growth (single highest-leverage lever feeding 5+ downstream items); the #172 ship-gate target.

**\*\*Serving:\*\*** explicitly deferred to the separate scan ? only the two overlapping cheap items (DOC\_STATUS refresh + the in-flight Ops Agent PR) surfaced.

**\*\*Stale flags consolidated\*\*** across both repos: T12/DRAFT understatement, REQUEST\_ACCESS\_FORM P2, catalog-location drift, DOC\_STATUS lagging merged M5/M6/M8, khelsutra README still calling DRY\_RUN\_CUJ "active", RALLY\_QUALITY §7 n=6 vs n=15, ROADMAP/COMMERCIALIZATION/BACKLOG stale dates and link rot.